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Math 201 Major Exam II Term 223

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The directional derivative of $f(x,y) = e^{-(x^2+y^2)}$ at $(0,0)$ in the direction of $\vec{v} = \langle 1,1 \rangle$ is

(a) 0

(b) -1

(c) 1

(d) 2

(e) -2

2. If $f(x,y) = x - y^2$, then the tangent line to the level curve $f(x,y) = 3$ at the point $(4,-1)$ is

(a) $7y = -2x + 1$

(b) $y = x + 1$

(c) $-7y = -x + 11$

(d) $2y = -x + 2$

(e) $y = -x + 3$

3. $\lim_{(x,y)\to(0,0)} e^{\left(\frac{1-\sec(x^2+y^2)}{x^2+y^2}\right)} =$

(a) ∞

(b) e

(c) $-\infty$

(d) 1

(e) 0

4. Let $f(x,y,z) = ye^x + x\ln z$, then $xf_{xz}f_{zx} + f_{zz} =$

(a) 1

(b) 0

(c) $\dfrac{x}{z^2}$

(d) $\dfrac{-x}{z^2}$

(e) $\dfrac{-x}{z}$

5. Let $w = xy^2 + yz^2 + zx^2$, $x = t^2$, $y = t^2 - 2s$, $z = s^2 - 2t$. The value of $\dfrac{\partial w}{\partial s}$ when $t = 1$, $s = 1$ is equal to

(a) 4

(b) 8

(c) 2

(d) 6

(e) 0

6. Let

$$f(x,y) = \begin{cases} \dfrac{3x^2y}{x^4+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

Which one of the following statements is FALSE?

(a) The domain of f is $\mathbb{R}^2$

(b) $f_x(0,0) = 0$

(c) f is not continuous at $(0,0)$

(d) f is not differentiable at $(0,0)$

(e) $\lim_{(x,y)\to(0,0)} f(x,y)$ exists

7. The domain of the function $f(x,y,z) = \dfrac{\sqrt{x^2+y^2+z^2-64}}{y}$ is

(a) The set of all points lying on or inside the sphere with center at the origin and radius 8 and are not on the xz-plane.

(b) The set of all points lying on or outside the sphere with center at the origin and radius 8 and are not on the x-axis.

(c) The set of all points lying on or inside the sphere with center at the origin and radius 4 and are not on the y-axis.

(d) The set of all points lying on or outside the sphere with center at the origin and radius 8 and are not on the xz-plane.

(e) The set of all points lying on or outside the sphere with center the origin and radius 4 and are not on the x-axis.

8. The distance between the two lines

$$L_1: \ -2x = y - 3 = 2z - 8 \quad \text{and} \quad L_2: \ 2x = 1 - y = 8 - 2z,$$

is equal to

(a) $\dfrac{4}{3}$

(b) $\dfrac{\sqrt{12}}{3}$

(c) $\dfrac{\sqrt{22}}{3}$

(d) $\dfrac{2}{3}$

(e) $\dfrac{\sqrt{66}}{3}$

9. The level curves of $f(x, y) = e^{1-x^2+y^2}$ are

(a) circles

(b) hyperbolas

(c) lines

(d) ellipses

(e) parabolas

10. The surface

$$4x^2 - 3y^2 + 12z^2 + 12 = 0$$

is

(a) a hyperboloid of two sheets with the x-axis as its axis

(b) a hyperboloid of one sheet with the y-axis as its axis

(c) a hyperboloid of two sheets with the y-axis as its axis

(d) a hyperboloid of one sheet with the z-axis as its axis

(e) a hyperboloid of one sheet with the x-axis as its axis

11. An equation of the surface that consists of all points equidistant from the point $(2, 2, 2)$ and the xy-plane is

(a) $\dfrac{(z-1)^2}{2} = \dfrac{(x-2)^2}{4} + \dfrac{(y-2)^2}{4}$

(b) $z = \dfrac{(x-2)^2}{4} + \dfrac{(y-2)^2}{4} + 1$

(c) $\dfrac{(z-2)^2}{4} = (x-2)^2 + (y-2)^2$

(d) $z = (x-2)^2 + (y-2)^2 + 2$

(e) $y = \dfrac{(x+2)^2}{4} + \dfrac{(z+2)^2}{4} + 2$

12. If (x, y) moves from the point $(2, 2)$ to the point $(2.04, 1.98)$, then, using differentials, we find that the change in $z = \sqrt{3x^2 + y^2}$ is approximately

(a) 0.03

(b) 0.06

(c) 0.02

(d) 0.05

(e) 0.04

13. The distance between the planes

$$2x - 3y + 6z = 6 \quad \text{and} \quad -4x + 6y - 12z = 2$$

is equal to

(a) 3

(b) 2

(c) 8

(d) 1

(e) 6

14. Let $f(x, y) = x^y$, then $f_y(e, 1) =$

(a) 0

(b) e^e

(c) e

(d) 1

(e) e^2

15. $\lim_{(x,y)\to(0,1)} \tan^{-1}\left(\frac{x^2-1}{x^2+(y-1)^2}\right) =$

(a) 0

(b) $\frac{\pi}{2}$

(c) DNE

(d) $-\frac{\pi}{2}$

(e) $-\frac{\pi}{4}$

16. If the plane containing the point $(2, 1, 2)$ and the line given by

$$\frac{x-1}{3} = \frac{y-2}{4} = z+1$$

passes through the point $(-1, b, 1)$, then $b =$

(a) -2

(b) 1

(c) 2

(d) 3

(e) -3

17. If the temperature at the point (x, y) on a metal plate is $T(x, y) = \frac{x}{x^2+2y^2}$, the direction of the greatest increase in heat from the point $(1, -1)$ is

(a) $\left\langle \frac{1}{9}, \frac{-2}{9} \right\rangle$

(b) $\left\langle \frac{1}{25}, \frac{-4}{25} \right\rangle$

(c) $\left\langle \frac{1}{9}, \frac{4}{9} \right\rangle$

(d) $\left\langle \frac{4}{9}, \frac{1}{9} \right\rangle$

(e) $\left\langle \frac{2}{9}, \frac{-1}{9} \right\rangle$

18. If $x\cos 3y + x^3y^5 = 3x - e^{xy}$, then the value of $\dfrac{dy}{dx}$ at the point $(1, 0)$ is

(a) 2

(b) -1

(c) 1

(d) -2

(e) 0

19. If z is defined implicitly as a differentiable function of x and y by the equation

$$x\ln y + y^2z - z^2x = 10$$

then $\dfrac{\partial z}{\partial x} =$

(a) $\dfrac{z^2 - \ln y}{y^2 - 2xz}$

(b) $\dfrac{z^2 - \ln y}{y^2 + 2xz}$

(c) $\dfrac{y^2 - \ln y}{y^2 - 2xz}$

(d) $\dfrac{z^2 - x}{y^2 - 2xz}$

(e) $\dfrac{z^2 + \ln y}{y^2 + 2xz}$

20. If the radius and height of a right circular cylinder are respectively measured as 10 cm and 30 cm with possible errors ± 0.02 cm and ± 0.04 cm, respectively, then the error in calculating the volume of the cylinder is approximately

(a) $\pm 15\pi\ \text{cm}^3$

(b) $\pm 12\pi\ \text{cm}^3$

(c) $\pm 16\pi\ \text{cm}^3$

(d) $\pm 20\pi\ \text{cm}^3$

(e) $\pm 18\pi\ \text{cm}^3$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$\nabla f(0,0) = \mathbf{0}$, so $D_{\mathbf{u}} f = 0$.
2	**(d)**	$\nabla f(4,-1) = \langle 1, 2\rangle$: $(x-4) + 2(y+1) = 0$, i.e. $2y = -x + 2$.
3	**(d)**	Exponent $\to 0$ as $r = x^2 + y^2 \to 0$, so the limit is $e^0 = 1$.
4	**(b)**	$f_{xz} = f_{zx} = \frac{1}{z}$, $f_{zz} = -\frac{x}{z^2}$, so the sum is 0.
5	**(b)**	$w_s = w_y(-2) + w_z(2s) = (-1)(-2) + 3(2) = 8$.
6	**(e)**	Limit is 0 along $y = 0$ but $\frac{3}{2}$ along $y = x^2$, so the limit does not exist.
7	**(d)**	$x^2 + y^2 + z^2 \geq 64$ and $y \neq 0$.
8	**(b)**	Parallel lines, $d = \frac{\|PQ \times \mathbf{d}\|}{\|\mathbf{d}\|} = \sqrt{8/6} = \frac{\sqrt{12}}{3}$.
9	**(b)**	$y^2 - x^2 = \ln c - 1$: hyperbolas.
10	**(c)**	$-\frac{x^2}{3} + \frac{y^2}{4} - z^2 = 1$: two sheets, y-axis.
11	**(b)**	$(x-2)^2 + (y-2)^2 + (z-2)^2 = z^2$ gives $z = \frac{(x-2)^2 + (y-2)^2}{4} + 1$.
12	**(d)**	$dz = 1.5(0.04) + 0.5(-0.02) = 0.05$.
13	**(d)**	Second plane is $2x - 3y + 6z = -1$; distance $\frac{\|6+1\|}{7} = 1$.
14	**(c)**	$f_y = x^y \ln x$, so $f_y(e,1) = e \ln e = e$.
15	**(d)**	Fraction $\to -1/0^+ = -\infty$, so the limit is $-\frac{\pi}{2}$.
16	**(e)**	Normal $\langle 13, -8, -7\rangle$; $-26 - 8(b-2) - 14 = 0$ gives $b = -3$.
17	**(c)**	$\nabla T(1,-1) = \langle \frac{2y^2 - x^2}{9}, \frac{-4xy}{9}\rangle = \langle \frac{1}{9}, \frac{4}{9}\rangle$.
18	**(a)**	$F_x(1,0) = -2$, $F_y(1,0) = 1$, so $\frac{dy}{dx} = -\frac{F_x}{F_y} = 2$.
19	**(a)**	$F_x = \ln y - z^2$, $F_z = y^2 - 2xz$, so $z_x = \frac{z^2 - \ln y}{y^2 - 2xz}$.
20	**(c)**	$dV = 2\pi rh\, dr + \pi r^2 dh = 12\pi + 4\pi = \pm 16\pi$ cm^3.